A predictive latent dynamics model for extreme gust encounters

Toward model-based flow control

Forward physical closure of predictive vs reconstructive latents at Re = 5000

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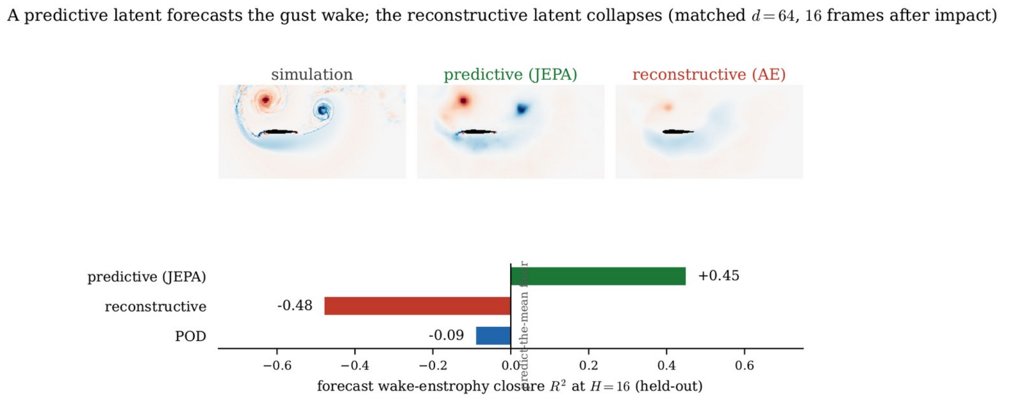
EUROMECH Colloquium · Data-driven active control in flows: from model-based to reinforcement learning

MOTIVATION

The control problem

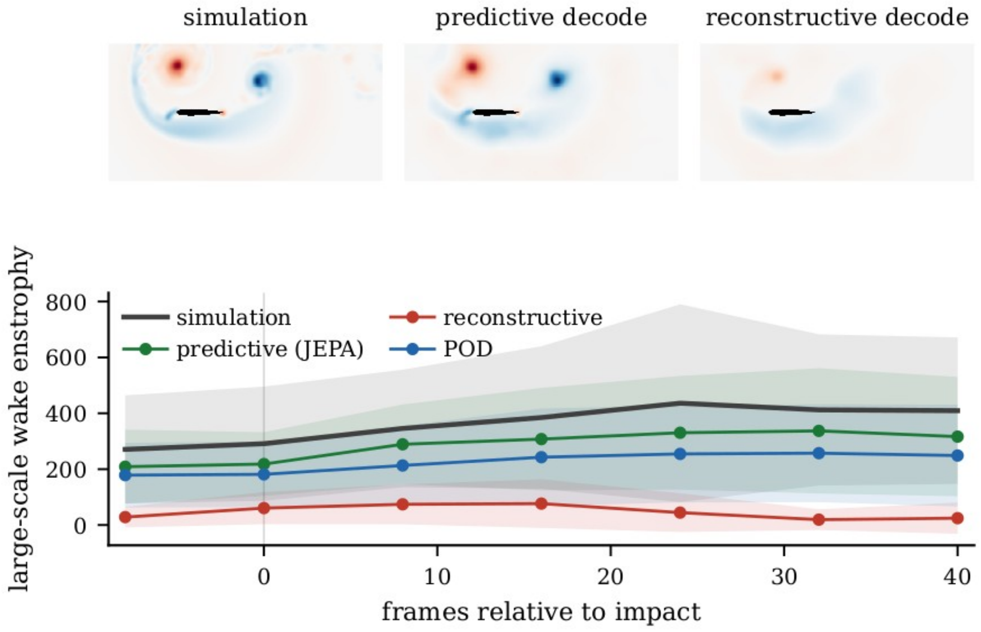
- Small and micro air vehicles increasingly fly where the **gust speed is comparable to or larger than the flight speed**, urban canyons, building and ship wakes
- Gust ratio **G > 1**: large, fast load transients that conventional gust models miss
- Active control needs a **fast, faithful forward model**, model-based control and RL world-models both rest on a learned reduced dynamics
- This talk: **what reduced state is worth planning against?**



Parametric vortex-gust encounters on a NACA 0012 at Re = 5000.

Why it is hard: it is the wake, not just the forces

- The load transient is built by **leading-edge-vortex (LEV)** formation, growth and shedding, plus the wake left behind
- The discriminating information lives in the **wake**, not only in the integrated forces C_L, C_D
- A model that tracks forces but loses the wake looks fine **instantaneously, yet fails under rollout**
- α = 14°, Re = 5000: massively separated, nonlinear coupling with the natural shedding



Large-scale wake / LEV organisation governs the transient.

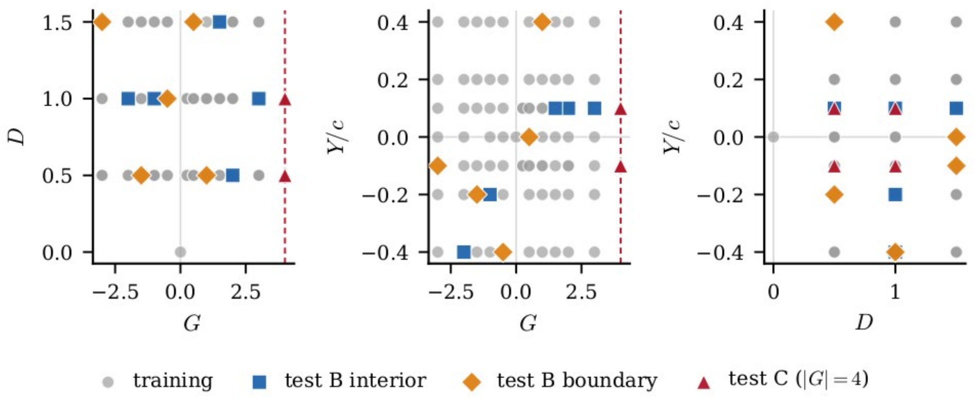
The opening for JEPA

- Most flow ROMs, POD/DMD, autoencoders, are trained to **reconstruct the field**
  - reconstruction fixes the latent only up to a diffeomorphism, so its geometry is not constrained to be predictable
- Alternative: learn a state that is **predictive by construction**, and plan against it

> **Which reduced state stays physically closed when propagated forward, so that every observable remains recoverable along the rollout?**

Data

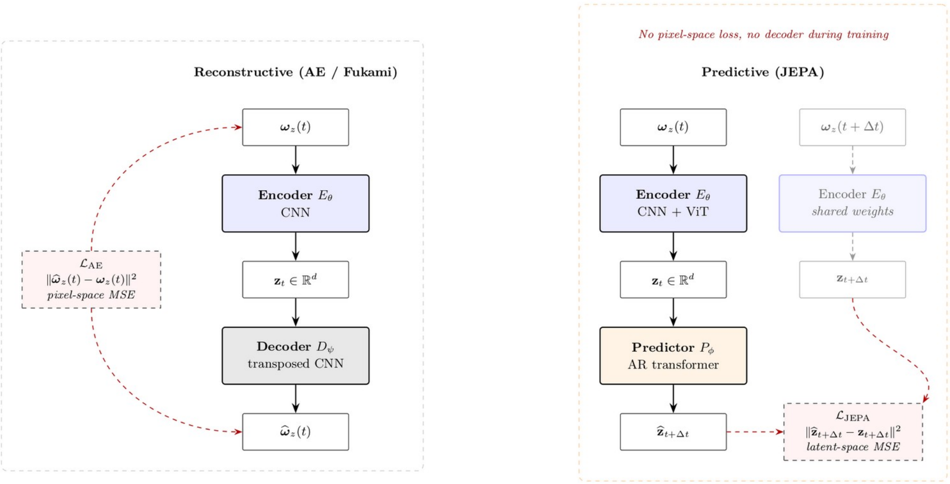
- **DNS** (SOD2D, no subgrid-scale model), NACA 0012, α = 14°, **Re = 5000**
- Taylor-vortex gusts parametrised by strength **G**, core diameter **D**, wall-normal offset **Y**
- 84 cases; impact-centred encounters; six observables: C_L, C_D, impulse I_y, **wake enstrophy**, ±circulation
- Held-out **test_b** (interpolation) and **test_c** (|G| = 4 extrapolation)



Training envelope in (G, D, Y); held-out interpolation and extrapolation.

What is a JEPA?

- **JEPA** = Joint-Embedding Predictive Architecture (LeCun's world-model program)
- Encoder x → z; a predictor advances **z** in time; the **loss lives in latent space, the field is never reconstructed** (an anti-collapse term replaces the decoder)
- So the encoder drops detail not predictive of the future, and the predictor gets no pixel-level shortcuts
- Lineage: I-JEPA / V-JEPA (image, video); from-pixels LeWM, LeJEPA, PLDM, so far only gridworld and toy visual tasks
- **Why for control:** a learned world-model, cheap latent rollout, keeps control-relevant dynamics, sensor-observable; the **first end-to-end JEPA on a parametric flow** (to our knowledge)

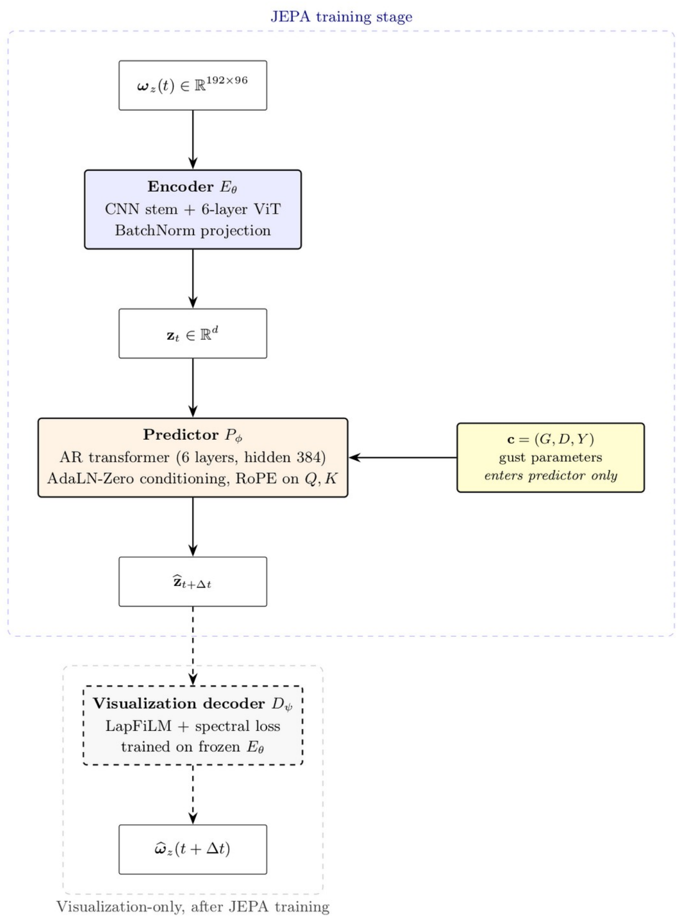


Reconstruct the field vs predict in latent space.

METHOD

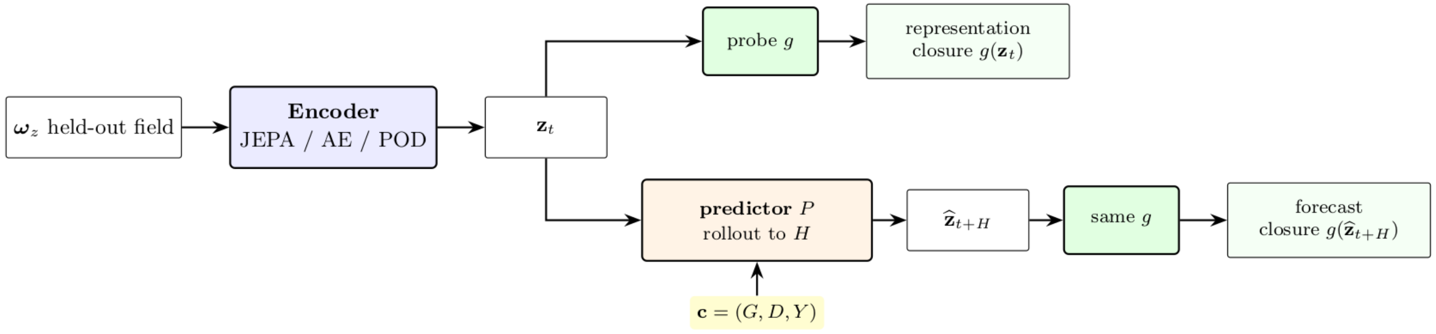
Our architecture for gust encounters

- **Encoder** (hybrid CNN + ViT) is **unconditional**: mid-plane vorticity ω_z → latent z (d = 32 / 64)
- **Predictor** (autoregressive transformer): the gust **c = (G, D, Y) enters the predictor only** (AdaLN, RoPE, causal mask)
- The predictor **is** the latent dynamics model, the object you plan or learn against
- A visualisation decoder is trained on the **frozen** encoder, never in the loss
- Compared at matched latent dimension vs a reconstructive AE (Fukami lineage) and POD



Forward-closure protocol

- **Forward closure:** roll the predictor recursively to **H = 16** frames after impact, then probe each observable from the predicted latent
- **Matched protocol:** same predictor architecture and same probe family, **trained / fitted separately per latent**, so differences are the encoder's
- **Conditioning-only floor:** KRR on (G, D, Y) alone, the latent must beat it to prove the state, not the parameters, does the work
- Reported with bootstrap CIs, 3 encoder seeds, 5-fold probe CV; held-out test_b / test_c

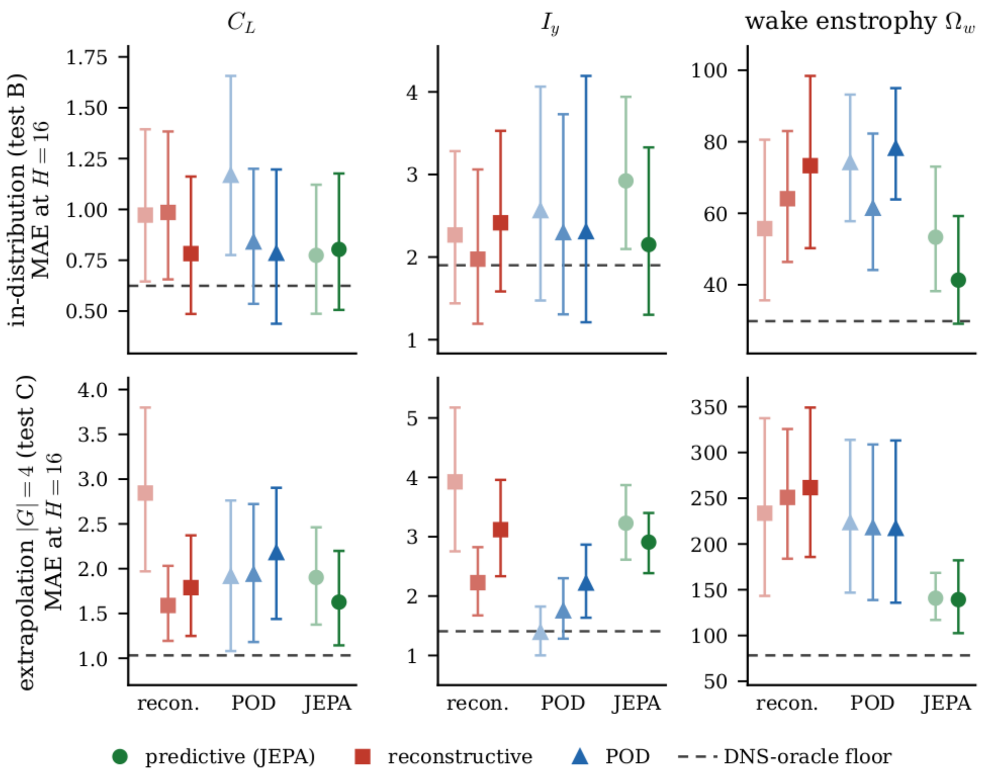


The predictor architecture, the conditioning $\mathbf{c}$, and the probe family g are identical across the three encoder families, each trained or fitted separately per family.

The same predictor and probe map every encoder family to the observables, in two modes.

Main result: forward closure

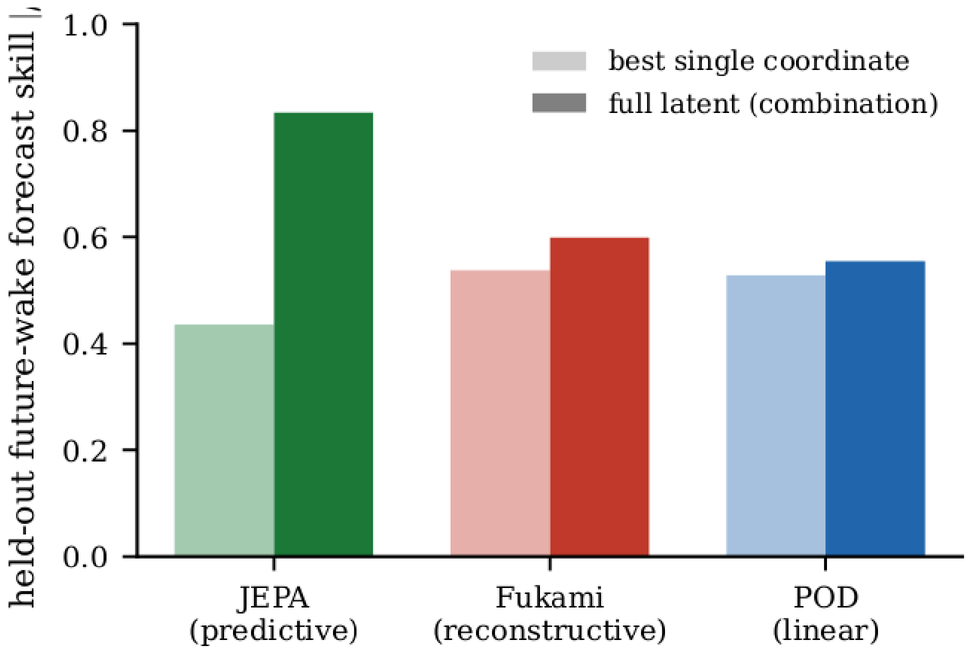
- Closure R^2 (rolled-out latent → observable, mean over 6): **JEPA 0.84 · Fukami 0.43 · POD 0.56**
- **Wake enstrophy** is the discriminator: **0.93** vs 0.28 (Fukami) / 0.37 (POD)
- Held-out MAE at H = 16: JEPA wake enstrophy **2.4× lower** than the AE, **3× lower** than POD
- POD stays competitive only on the integrated impulse I_y



Held-out MAE at H = 16 by observable (test B, test C); JEPA lowest on the wake.

It is the wake, not the forces

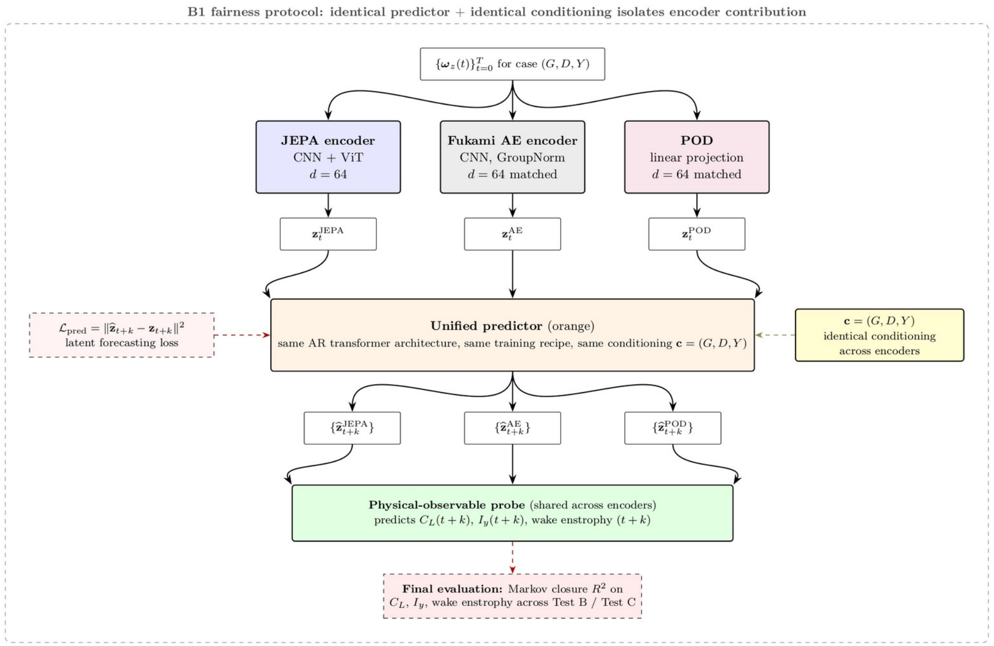
- **Forces** (C_L, C_D) are forecast **redundantly by every family**, many coordinates carry them
- Only the predictive latent carries a **distributed wake-forecast code** (rank corr 0.83; best single coordinate 0.44)
- Reconstructive / linear: best single coordinate ≈ the whole latent → **no collective wake structure**
- And it clears the **conditioning-only floor** → the latent, not the parameters, does the work



Wake forecast skill: full latent (dark) vs best single coordinate (light).

Controls: objective and supervision, not architecture

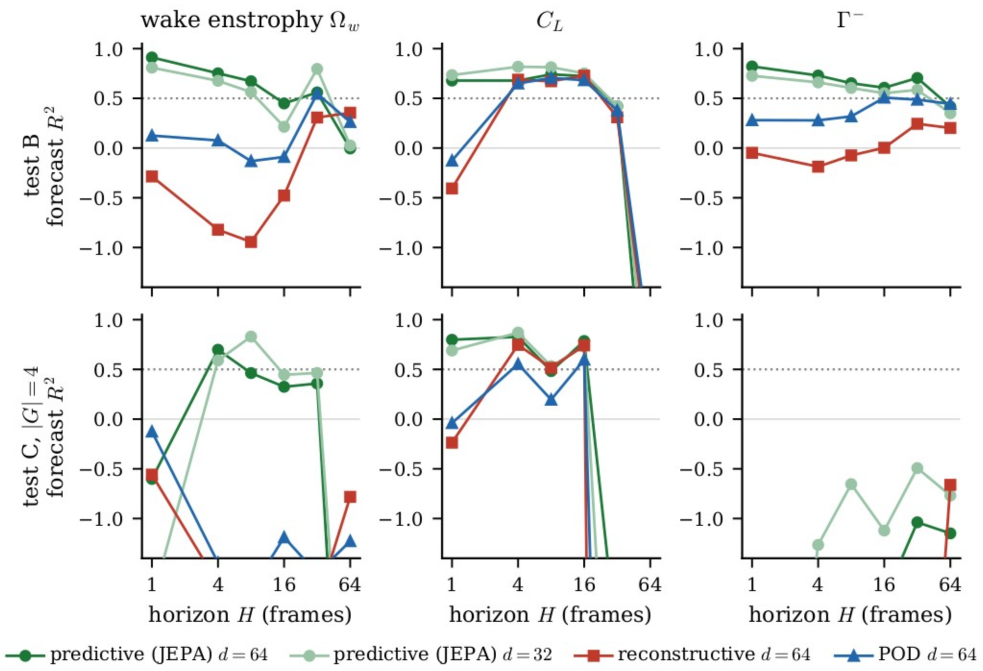
- **2 × 2:** objective {predictive, reconstructive} × architecture {CNN, CNN+ViT}, auxiliary heads matched
- Predictive beats reconstructive at **both** architectures (wake R^2 0.46 / 0.45 vs 0.16 / 0.29) → **not the ViT**
- Remove the wake head from the predictive model → wake closure **collapses below floor (R^2 = −1.03)**
- → the gain needs the **predictive objective and wake supervision together**



Objective × architecture controls, three seeds per cell.

Diagnostic 1: latent drift

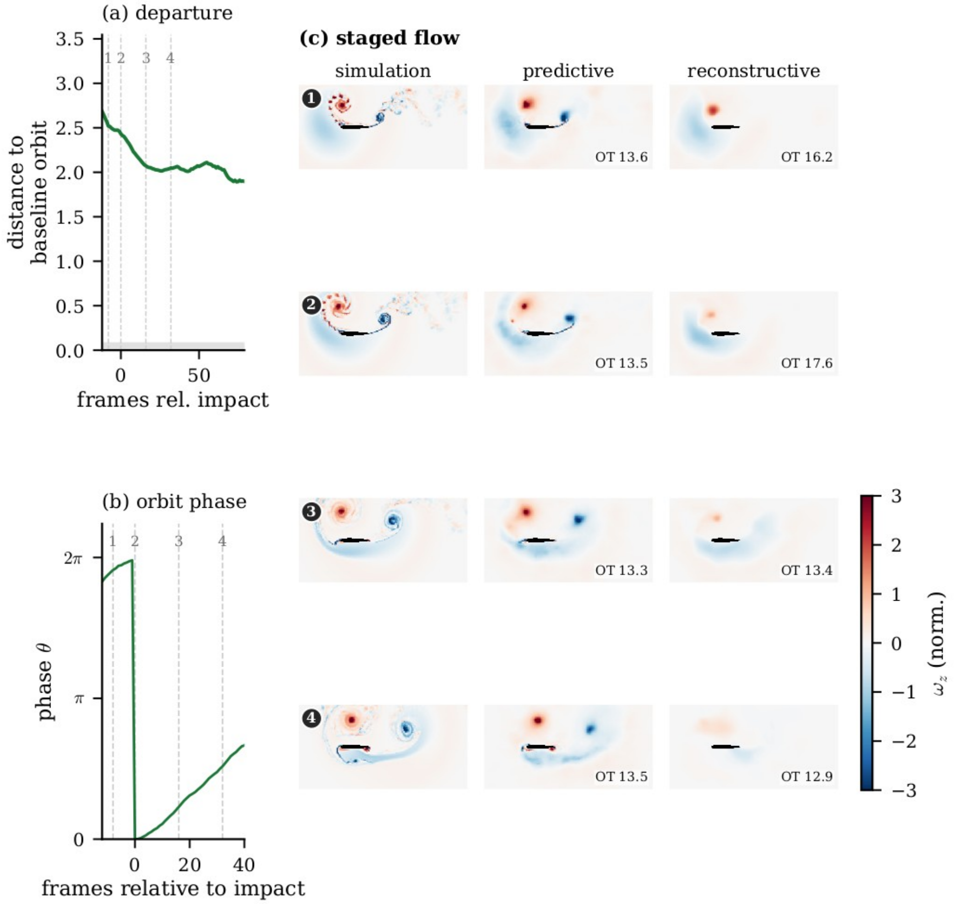
- **Why this question:** a planner / RL agent queries the model at states reached by its **own rollout**, one-step error is not enough
- **Mahalanobis distance:** covariance-aware distance from a reference cloud, d = sqrt((z−μ)' S^-1 (z−μ)); ≈ 1 is one standard-deviation unit
- We compare the **rolled-out** latent to the distribution of **DNS-encoded** latents (ratio = rollout / encoded)
- **Result:** the reconstructive rollout drifts **~10× off-manifold (9.9)**; JEPA 0.85 and POD 0.81 stay inside



Forecast skill vs rollout horizon, held-out test_b.

Diagnostic 2: topology of the encounter

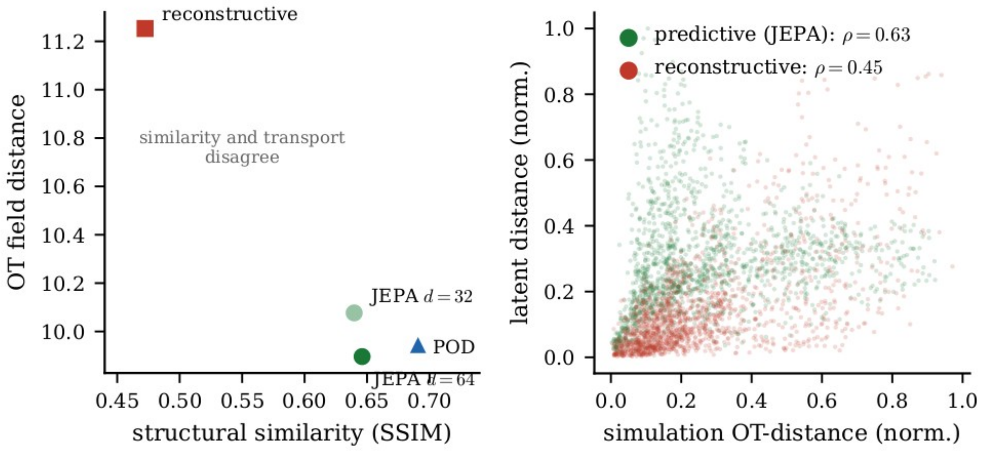
- **Why:** shedding and gust encounters are **cyclic**; a faithful dynamics model should keep the trajectory **one closed loop** under rollout
- **Persistent homology:** counts topological loops (1-cycles) and how long they persist as a scale parameter grows; one long-lived cycle = a clean loop, many short-lived ones = fragmentation
- **Result:** the predictive latent traces a **single persistent cycle**; the reconstructive latent **fragments**
- → the predictive state is a coherent object to integrate forward



The encounter is a single cycle in the predictive latent.

Diagnostic 3: transport geometry

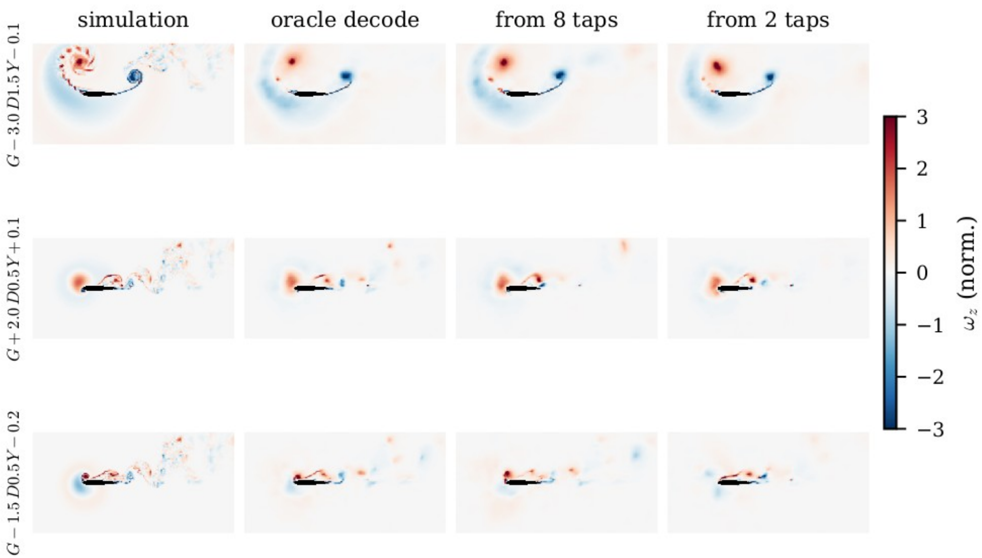
- **Why:** Euclidean / pixel distance ignores **where structures move**; control cares about advection of the LEV and shear layer
- **Optimal-transport (OT) distance:** the least cost to rearrange one vorticity field into another, sensitive to transport
- We Spearman-correlate the **latent** distance matrix with the **OT** distance matrix along each encounter
- **Result:** JEPA **0.63** vs Fukami 0.45, order-preservation, not an isometry; latent steps mean physically sensible field changes



Latent vs OT distance alignment (per-encounter Shepard).

Physical space: the large-scale wake is recovered

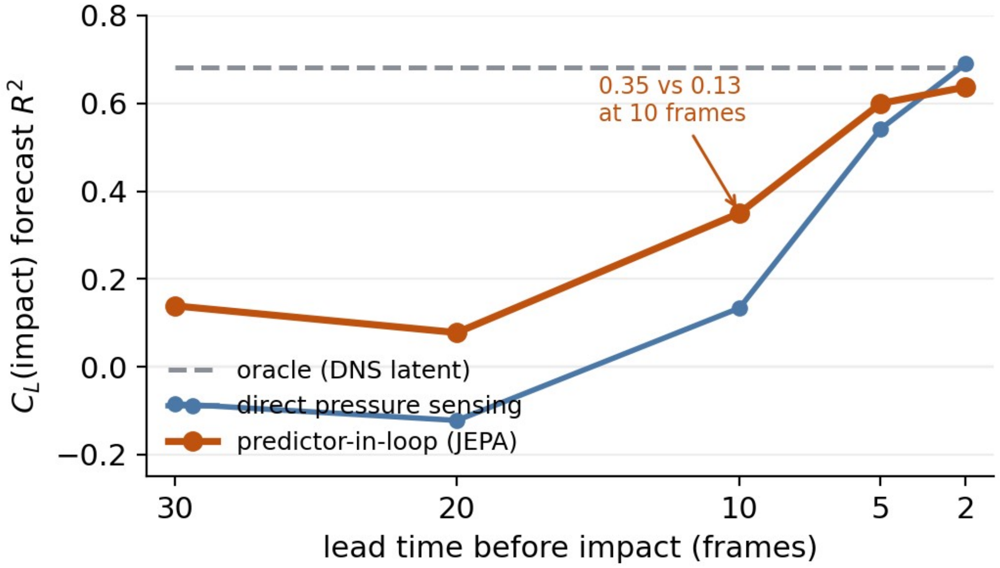
- Decode the latents back to fields (frozen-encoder diagnostic, never in the loss)
- The predictive decode tracks the **large-scale LEV / wake**: Pearson **0.89** at impact, **0.91** at H = 16
- The reconstructive decode is sharper pixel-wise but **loses the transported wake** under rollout
- This is the explicit trade the predictive objective makes



Large-scale wake recovery, predictive vs reconstructive.

Control relevance: observable and forecastable

- The predictive state is **observable from sparse wall pressure**: (G, D) recoverable from the latent ($R^2 \approx 0.46$ / 0.80 on test_b)
- **Lead-time impact-C_L:** predictor-in-loop **R^2 = 0.35 at 10 frames pre-impact**, vs 0.13 from direct pressure sensing (oracle 0.68)
- The reconstructive latent's oracle is **negative**, a representation failure, not a probe failure
- A deployment-relevant **state estimate plus short-horizon forecast**: the ingredients a controller consumes



Inferring impact-C_L from K = 8 wall-pressure sensors (test_b): the predictor-rolled latent beats direct sensing well before impact.

Takeaways

- A **predictive objective + wake supervision** yields a latent dynamics model that is:
  - **forward-closed** (wake R^2 0.75 representational / 0.84 mean), **on-manifold** under rollout, a **single cycle**, **transport-aligned**
  - **compact**, $d = 32 \approx d = 64$, participation ratio ≈ 1.7
  - **observable** from sparse wall pressure
- A **conditional forward-closure model**, a substrate for **model-based control and RL world-models**
- The discriminator throughout is the **wake**, which force-only and reconstruction-only states lose

Outlook

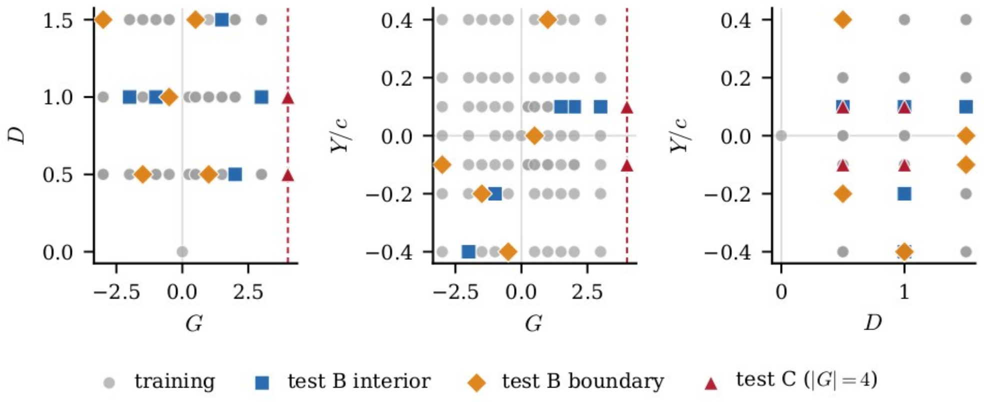
- Replace the gust conditioning channel with an **actuation channel**, the same forward-closure machinery then applies to control inputs
- **Closed-loop** model-based control / RL world-models built on the latent
- **3D observability** at strong gusts ($|G| = 4$ is a mid-plane-slice limit)
- The predictive-latent recipe carried to **other parametric flows**

Backup slides

Supporting detail and ablations

Dataset and protocol

- Partition **v2** (locked, sha256-anchored): 84 cases
- **226** train encounters · **42** test_b · **24** test_c (|G| = 4)
- ω_z pipeline v1: spatial mask + p99.99 clip + 3σ scale (train_std 3.55), no mean shift (preserves vorticity antisymmetry)
- Reporting: bootstrap n = 2000 CIs · 3 encoder seeds · 5-fold probe CV



Training envelope in (G, D, Y).

Matched-capacity: d = 32 vs d = 64

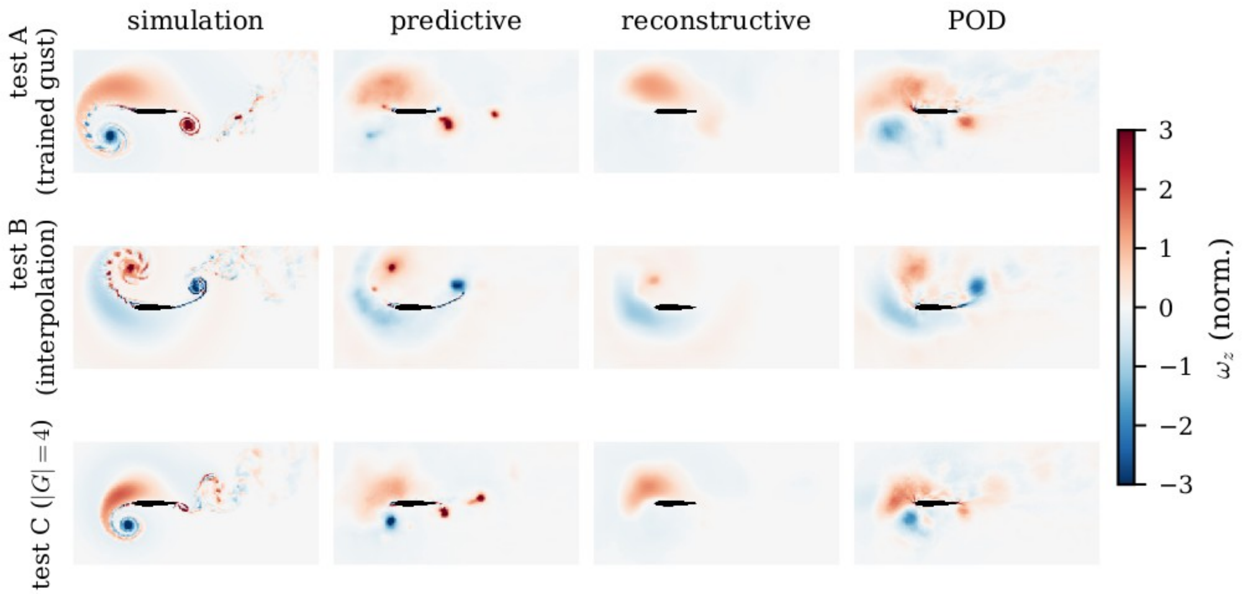
- Halving the latent leaves the representation and **every mechanism diagnostic** intact
  - representational wake R^2 0.74 (d=32) vs 0.75 (d=64); drift ratio 0.86 vs 0.85; OT 0.61 vs 0.63
- Cost is **in-distribution forecast sharpness**: H = 16 wake closure 0.45 → 0.21
- Participation ratio ≈ 1.7, leading PC ≈ ¾ of variance → the effective dimension is a handful

Seed variance

- Three encoder seeds per cell; wake-closure means reported ± standard deviation
- The reconstructive CNN+ViT cell has **large seed variance (±0.27)**, consistent with the drift mechanism: without a forward-predictable geometry, closure is unstable across initialisations
- Predictive cells are tight (±0.03–0.06)

Decoded reconstructions

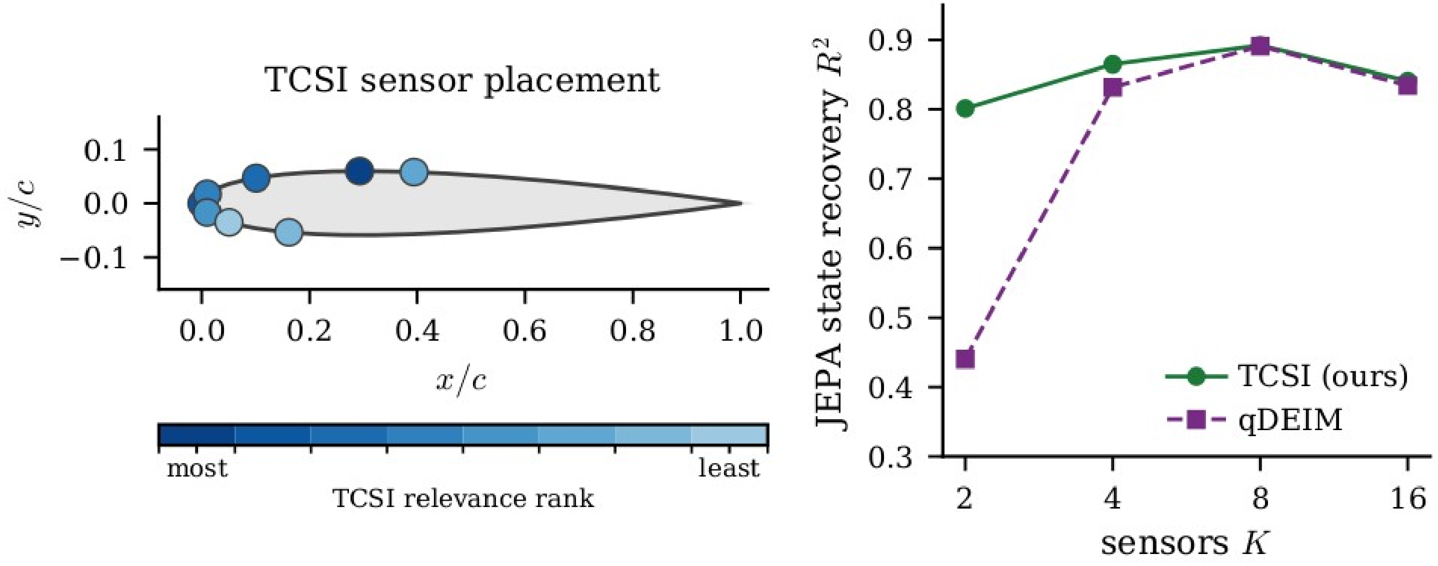
- Decoded fields: predictive vs reconstructive vs POD vs DNS
- The predictive decode is **blurrier pixel-wise** but preserves the **transported large-scale wake**; the reconstructive decode is sharp yet drifts under rollout



Held-out reconstructions; the predictive objective trades pixel sharpness for transported structure.

Sparse sensor placement

- Sparse wall-pressure sensors selected by **TCSI / qDEIM** vs uniform
- Pressure → latent map: KernelRidge(RBF) on the K-sensor × pre-impact-window vector
- K = 2 / 4 / 8 / 16; the LSTM/KRR comparison is 5-fold-CV selected to guard small-sample overfitting

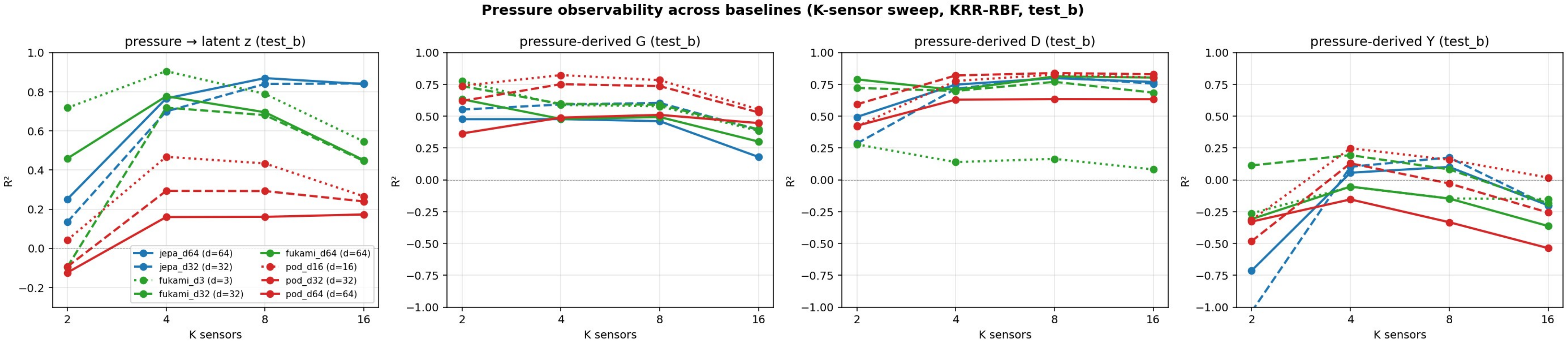


Optimal sparse sensor placement on the airfoil surface.

BACKUP

Parameter observability from the latent

- $z \to (G, D, Y)$ probe R^2 from the rolled-out latent, K = 8 pre-impact sensors
- test_b: **G 0.46 · D 0.80 · Y 0.10**, gust impulse and depth recoverable; cross-stream offset Y is marginal
- test_c (|G| = 4): negative, the 3D observability boundary of a single mid-plane slice



Gust parameters implicit in the latent on held-out cases.

SSIM convention

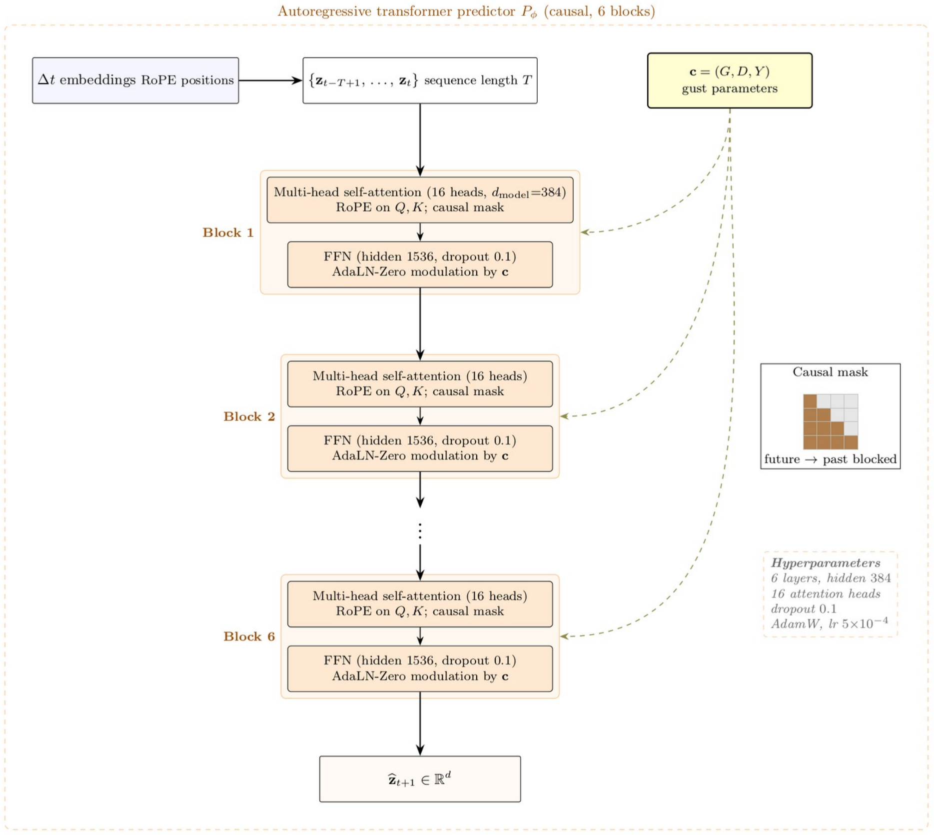
- Reconstruction SSIM uses the **Wang convention** ($K1 = 0.01$, $K2 = 0.03$) on pipeline-normalised ω
- Data range **$L = 2 \cdot$ global p99.9(|target|) $\approx$ 8.31** (split v2)
- Decoder test_a SSIM $\approx$ **0.71** at this convention

Conditioning-only floor

- KRR-RBF on (G, D, Y) → observable at impact; train / test_b / test_c R^2
  - on train, (G, D, Y) interpolates C_L and C_D well (226 points in 3-D)
- On **test_b** the floor **collapses** across observables; on **test_c** it is negative for 5 of 6
- → JEPA's generalisation is **not** explained by the conditioning alone

Predictor detail

- Predictor: 6-layer autoregressive transformer, hidden 384, 16 heads, dropout 0.1
- **AdaLN-Zero** conditioning on (G, D, Y, φ_t); **RoPE** temporal positions; **causal** mask
- Training: teacher forcing + scheduled-sampling rollout (H_roll = 8); **SIGReg** anti-collapse (VICReg fallback)



Phase–amplitude reading

- Phase–amplitude decomposition of the encounter cycle in the predictive latent
- Connects the predictor to the **sensitivity-function control** designed for these flows
- An actuation channel would enter where the gust does, the predictor conditioning